Evidence-verifying agents for discharge coordination

Healthcare Agentic AI
Maria Vargas-Ledesma | Prepared for Dr. Khem Poudel

The project is a working local research prototype using synthetic patient cases. It combines local MedGemma output with deterministic evidence checks, supervised review, and reproducible learning experiments. This package supports a technical and research-direction review; it is not a claim of clinical validation or a finished conference paper.

What is ready to review

Area	Implemented behavior
Agentic workflow	Bounded inspect, scan, verify and route steps; source-quality clarification; retained rejected proposals; hash-linked tool traces.
Safety and provenance	Exact proposal/evidence matching, approval controls, immutable source snapshots, content digests and optional HMAC signatures.
Backend and database	Transactional persistence of full runs, recommendations and governance; run-bound append-only reviews; retry-safe request IDs; history restoration.
Learning experiments	Trainable contrastive case encoder and actual tabular Q-learning, with baselines, ablations, saved weights and explicit limitations.

Current status

FastAPI, Ollama and PostgreSQL were tested together. The interface provides Agent Research, Run History and Human Review. All 56 unittest tests pass, including isolated persistence contracts. The frontend production build and lint checks pass. A live MedGemma medication-review case passed proposal verification and retained the human-approval requirement.

No clinical action is executed. Model-generated rationale remains unverified prose even when its task metadata passes the structural verifier.

From generated proposals to durable review records

The previous backend saved cases, findings and tasks, but human decisions existed only in JSONL files. The update makes PostgreSQL the durable source of truth for each saved proposal and review.

Storage	What it preserves
Existing four tables	Synthetic cases, separate pipeline executions, detected barriers and detected tasks.
run_artifacts	A full response snapshot per new run, with input and payload hashes.
recommendation_records	Accepted proposals, uniquely identified within their saved run.
governance_records	Policy and approval outcomes tied to the saved recommendation by a foreign key.
human_review_records	Append-only decisions with a run ID, recommendation ID, reviewer label and retry-safe request ID.
schema_migrations	Applied revision and schema checksum; additive initialization is safe to rerun.

Consistency and review controls

Case upsert, execution, findings, tasks, proposals, governance and full artifact are committed in one transaction. A failure rolls back the entire save. Review requests must match the saved case, task and action. The backend checks the immutable artifact and source provenance, and rejects attempts to approve blocked or unpermitted recommendations.

Identical retries return the original review. Reusing the same request ID with changed content is rejected. A failed secondary JSONL write does not erase a committed database review. Reviewer identity is still a self-reported label; authentication and role-based authorization remain future work.

Migration and live checks

Revision **2026_11_review_storage_v1** added five tables and a history index. The original 7 case records, 14 runs, 28 barriers and 28 tasks were backed up and verified unchanged immediately after migration. Seven new deterministic reference runs matched their expected IDs. Simultaneous duplicate review submissions created one record; forged action metadata returned HTTP 409.

Older runs remain explicit legacy summaries. Full historical model responses were not invented or backfilled. Database backups and credentials are excluded from this review package.

Results that support the next research decision

The current corpus contains 240 assistant-authored combinations of existing synthetic scenarios. Representation learning uses 144 training, 48 validation and 48 test cases with no exact feature-group overlap. Shared generator assumptions remain; this is not an independently annotated clinical holdout.

Safety regression

2,160 mutated proposal inputs were flagged across nine mutation types, while all 240 clean reference proposal sets passed structural checks. This measures hand-authored metadata/evidence consistency tests. It does not measure comprehensive prompt-injection resistance, model jailbreak success or clinical safety.

Representation learning

Nearest-case retrieval	Barrier Jaccard	Task Jaccard
Learned contrastive encoder	0.479	0.484
Untrained random projection	0.481	0.481
Status-token similarity	0.560	0.560

Higher is better. The learned encoder does not outperform the simple token baseline. Its 12-dimensional weights are trained without answer labels; labels are used only to score retrieval. The experiment concerns structured case status, not images or a pretrained visual representation.

Reinforcement learning

A four-category toy workflow models inspect, verify and routing choices. Q-learning uses five seeds and 3,000 training episodes per training condition. Evaluation uses 400 balanced rollouts per seed and condition, with shielded/unshielded training and evaluation plus random and rule policies.

Both learned and rule policies reach 100% routing success in this small environment. Random-policy success is 0.5% without a shield and 74.4% with one. The shield prevents unsafe applied simulator actions while retaining unsafe proposal counts. These results demonstrate the implementation and the effect of the shield, not a benefit of learning over rules or transfer to clinical workflows.

Research claim boundary

The implemented components create a reproducible research foundation. Originality, clinical validity, a substantive vision contribution, and a competitive conference result have not been established. The next study should test a focused hypothesis on independently reviewed cases and stronger baselines.

A short demonstration and decisions for discussion

Step	What to inspect
1. Readiness	Open /ready or the API docs. Confirm database schema and the configured Ollama model are available.
2. Saved live run	Open Run History and load CASE-007 with model medgemma1.5:4b. Inspect barriers, the exact source field, model output and approval requirement.
3. Human review	Inspect the review queue. A flagged validation record is explicitly labelled automated; it is not a professor or clinician decision. New decisions stay bound to their saved run.
4. Agent Research	Inspect the tool trace, rejected-output handling, representation baselines and a clinical routing simulation.
5. Reproducibility	Review the frozen report, weights, split assignments and test evidence in this package.

Commands

```
python -m db.migrate
python -m scripts.review_check --live
python -m uvicorn api.main:app --reload
```

Frontend: cd frontend, then npm install and npm run dev. A new machine also needs PostgreSQL, the project Python dependencies and the configured Ollama model. The supplied .env.example contains placeholders; actual credentials are not included. Clean-room dependency installation has not been tested.

Questions for Dr. Poudel

Which contribution should lead the paper: evidence verification, supervised agent workflow, or a benchmark of conflicting sources? What independent annotation and domain review are needed? Is the current structured-data scope appropriate for the target venue? Should the next learning experiment focus on stronger representations or a richer routing environment where rules are a meaningful comparator?

Method references

Watkins and Dayan, Q-learning (1992): www.gatsby.ucl.ac.uk/~dayan/papers/cjch.pdf
Hadsell, Chopra and LeCun, Dimensionality Reduction by Learning an Invariant Mapping (CVPR 2006): yann.lecun.org/exdb/publis/pdf/hadsell-chopra-lecun-06.pdf
OWASP LLM Prompt Injection Prevention Cheat Sheet: cheatsheetseries.owasp.org

Detailed implementation: docs/backend_storage.md and docs/research_alignment.md. This brief was prepared with AI assistance; its measurements come from retained local test and experiment outputs.